

S4 — Biodiversity and Natural Resources

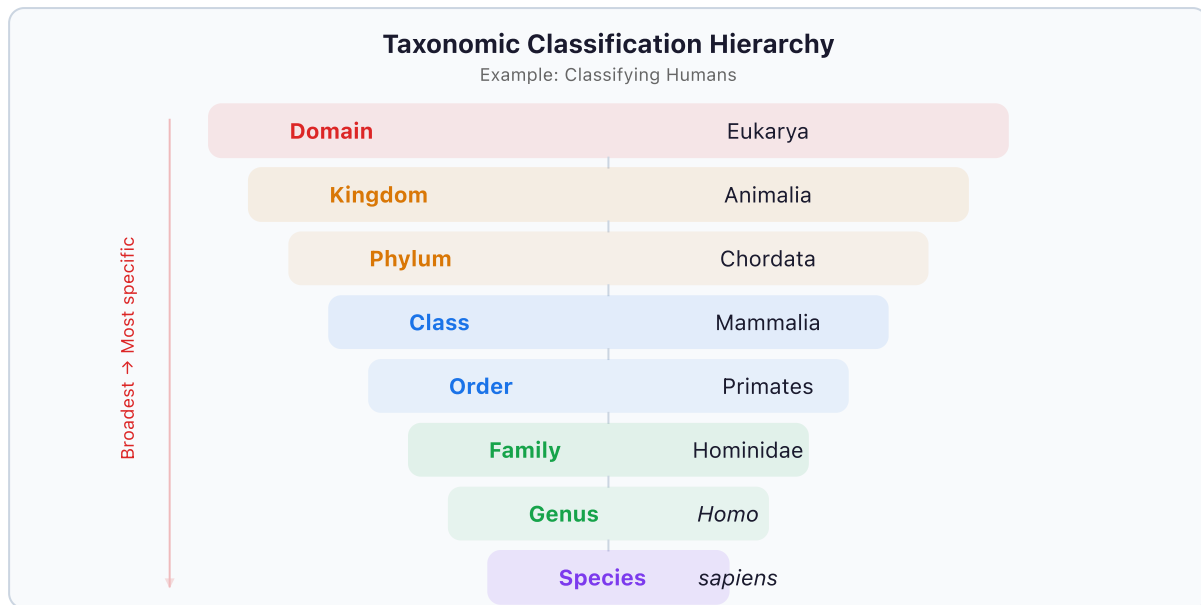
4.1 Classification and Taxonomy

The Hierarchy of Classification

Classification: The organisation of living organisms into groups based on shared features and evolutionary relationships. The hierarchical system arranges organisms from broad groups into increasingly specific categories.

The taxonomic hierarchy (from broadest to most specific):

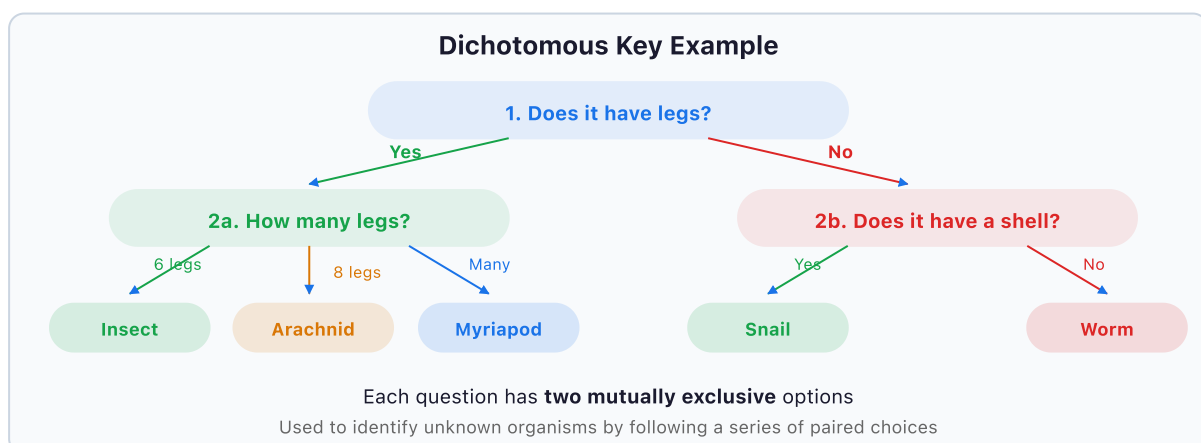
1. **Domain** — Bacteria, Archaea, Eukarya (the three-domain system proposed by Carl Woese based on rRNA analysis)
2. **Kingdom** — e.g., Animalia, Plantae, Fungi, Protocista, Prokaryotae
3. **Phylum** — e.g., Chordata (animals with a notochord/backbone)
4. **Class** — e.g., Mammalia
5. **Order** — e.g., Primates
6. **Family** — e.g., Hominidae
7. **Genus** — e.g., Homo
8. **Species** — e.g., sapiens



Species: A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other groups. The biological species concept is widely used but has limitations (does not apply to asexual organisms, fossils, or organisms that hybridise).

Binomial Nomenclature

Every species is given a two-part Latin name (binomial): the genus name (capitalised) followed by the species name (lowercase), both written in italics (or underlined if handwritten). For example, *Homo sapiens*, *Escherichia coli*. This system was developed by Carl Linnaeus and provides a universal naming system understood by scientists worldwide regardless of language.



Phylogenetics

Modern classification increasingly relies on **molecular phylogenetics** — comparing DNA sequences, RNA sequences (particularly ribosomal RNA, which evolves slowly and is found in all organisms), or amino acid sequences to determine evolutionary relationships. Organisms that share more similar sequences diverged more recently and are more closely related. The number of differences between sequences can be used as a **molecular clock** to estimate the time since two species diverged, assuming a roughly constant mutation rate.

Phylogenetic trees (cladograms) show evolutionary relationships, with branch points (nodes) representing common ancestors and branch lengths sometimes indicating evolutionary distance. Molecular evidence has led to reclassification of many organisms, sometimes contradicting traditional morphological classification. For example, molecular evidence showed that fungi are more closely related to animals than to plants, and led Carl Woese to propose the three-domain system (Bacteria, Archaea, Eukarya) based on ribosomal RNA sequences, replacing the older five-kingdom system.

Critical evaluation by the scientific community: New taxonomic groupings are accepted only after new data have been critically evaluated by other scientists. Findings are published in scientific journals after **peer review** (checking by independent experts), presented and discussed at conferences, and other scientists then repeat the work or test the conclusions with further data. Woese's proposal was initially met with scepticism and became widely accepted only after other scientists' molecular data supported it. 9BN0 names the three domains, based on molecular phylogeny, as **Bacteria, Archaea and Eukaryota**.

Advantages of molecular classification over morphological classification: It is more objective (based on quantifiable sequence data rather than subjective judgements about physical features); it can be applied to organisms with few observable features (e.g., bacteria); it reveals evolutionary relationships that convergent evolution may obscure (organisms may look similar due to similar environments, not shared ancestry); and it can resolve relationships at all taxonomic levels.

Exam Tip: Be prepared to interpret simple phylogenetic trees. The most closely related species share the most recent common ancestor (the nearest branch point). The length of branches may represent evolutionary time or the number of genetic changes. Remember that analogous structures (similar function, different evolutionary origin, e.g., wings of insects and birds) arise from convergent evolution and do not indicate close relationship, whereas homologous structures (same evolutionary origin, may have different functions, e.g., pentadactyl limb) indicate common ancestry.

4.2 Biodiversity

Levels of Biodiversity

Biodiversity exists at three levels:

- **Genetic diversity:** The variety of alleles within a species or population. Greater genetic diversity makes populations more resilient to environmental change.
- **Species diversity:** The number of different species (species richness) and the relative abundance of each species (species evenness) in an area.
- **Ecosystem diversity:** The variety of different ecosystems in a region (e.g., woodland, wetland, grassland, coral reef).

Biodiversity: The variety of living organisms in an area, including the number of different species and the genetic diversity within each species.

Endemism: The state of a species being found only in one particular geographical area and nowhere else in the world (e.g. lemurs are endemic to Madagascar).

Measuring Biodiversity Within a Habitat and Within a Species

Species richness is the number of different species in a habitat. It is a simple way to measure biodiversity within a habitat, but it does not take into account how many individuals of each species there are.

Genetic diversity within a species can be measured by calculating the **heterozygosity index (H)** for a gene locus:

$$H = \text{number of heterozygotes} / \text{number of individuals in the population}$$

For example, if 36 out of 120 individuals are heterozygous at a locus, $H = 36 / 120 = 0.3$. A higher value of H means greater genetic diversity. H is often calculated for several loci and averaged.

Measuring Biodiversity — Index of Diversity

Index of diversity (D): A quantitative measure used to compare the biodiversity of different habitats. It takes into account both the number of species and the number of individuals of each

species. The higher the value of D , the greater the diversity.

$$D = N(N - 1) / \sum n(n - 1)$$

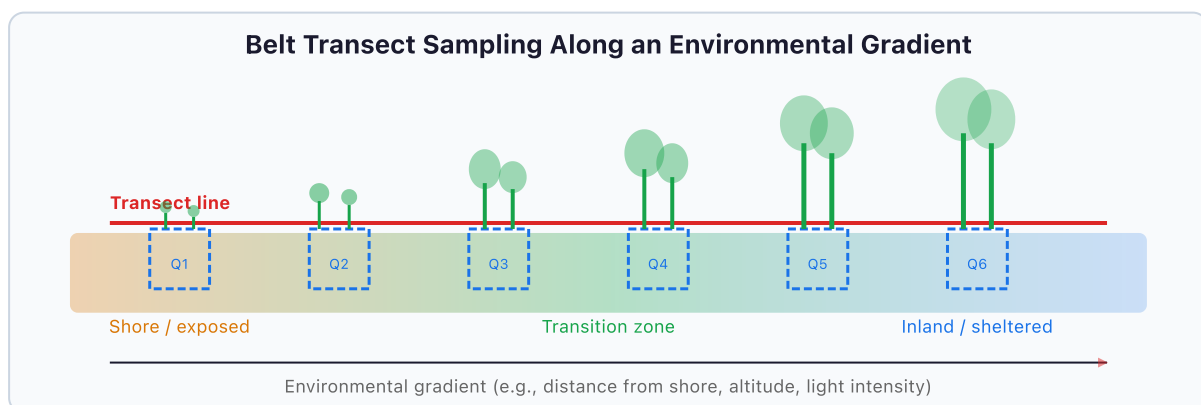
Where N = total number of organisms of all species, and n = number of organisms of each species. A habitat dominated by one species has low diversity (D close to 1, the minimum value), while a habitat with many species in similar numbers has a high value of D .

Exam Tip: 9BN0 uses $D = N(N-1)/\sum n(n-1)$. Work out $n(n-1)$ for each species, add them to get $\sum n(n-1)$, calculate $N(N-1)$ for the total, then divide. Example: 10, 5, 3 and 2 individuals of four species give $N = 20$, so $N(N-1) = 380$ and $\sum n(n-1) = 90 + 20 + 6 + 2 = 118$, giving $D = 380 / 118 = 3.22$. Show your working clearly.

Sampling Techniques

To measure biodiversity in the field, sampling techniques must be used because it is usually impractical to count every organism:

- **Quadrats:** Square frames placed randomly to sample sessile organisms (plants, slow-moving invertebrates). Record species frequency, percentage cover, or density. Random sampling using coordinates from a random number generator reduces bias.
- **Transects:** A line or belt transect is used to study changes in community composition along an environmental gradient (e.g., from shore to inland).
- **Mark-release-recapture (extension — not required by 9BN0):** Used for mobile animals. Capture a sample, mark them, release, then recapture. Population size is estimated using the Lincoln index.



$$\text{Population estimate} = (\text{number in first capture} \times \text{number in second capture}) / \text{number of marked recaptures}$$

Niche and Adaptation

Niche: The role of a species within its habitat — how it uses the biotic and abiotic resources of its environment and how it interacts with other organisms. Two species cannot occupy exactly the same niche in a habitat for long, because one will outcompete the other.

Organisms are **adapted** to their niche. Adaptations can be:

- **Behavioural:** The way an organism acts, e.g. migration, nocturnal activity to avoid heat, courtship displays, animals huddling together to keep warm.
- **Physiological:** Processes inside the organism, e.g. hibernation (lowering metabolic rate), desert mammals such as the kangaroo rat producing very concentrated urine to conserve water.
- **Anatomical:** Structural features, e.g. thick fur and a layer of blubber for insulation, a streamlined body shape, rolled leaves with sunken stomata in marram grass to reduce water loss.

Natural Selection

1. There is genetic variation within a population; new alleles arise by mutation.
2. A selection pressure (e.g. predation, disease, competition, a change in the environment) acts on the population.
3. Individuals with alleles that give an advantageous phenotype are more likely to survive and reproduce.
4. They pass these advantageous alleles on to their offspring.
5. Over many generations, the frequency of the advantageous alleles in the population increases, so the population becomes adapted to its environment.

Evolution is the change in allele frequencies in a population over time; natural selection is one of the main mechanisms by which it happens, and the accumulation of such changes over long periods can produce new species.

The Hardy–Weinberg Equation

For a gene with two alleles, where p = frequency of the dominant allele and q = frequency of the recessive allele:

$$p + q = 1 \quad p^2 + 2pq + q^2 = 1$$

p^2 = frequency of homozygous dominant individuals, $2pq$ = frequency of heterozygotes, and q^2 = frequency of homozygous recessive individuals. The equation predicts that allele frequencies stay constant from one generation to the next if the population is large, mating is random, and there is no mutation, no migration and no selection. Comparing the observed genotype or allele frequencies with those predicted, or calculating allele frequencies in the same population at different times, shows whether a change in allele frequency (evolution) is occurring.

Worked example: 1 in 100 individuals shows a recessive phenotype, so $q^2 = 0.01$ and $q = 0.1$. Then $p = 1 - 0.1 = 0.9$, and the frequency of heterozygotes = $2pq = 2 \times 0.9 \times 0.1 = 0.18$ (18%).

Reproductive Isolation and Speciation

If two populations of a species become **reproductively isolated**, gene flow between them stops. Isolation may be geographical (e.g. separated by a mountain range or sea), ecological (living in different habitats in the same area), seasonal/temporal (breeding at different times), behavioural (different courtship behaviour), or mechanical (incompatible reproductive organs). In each population, different mutations arise and different selection pressures act, so different genetic information (different allele frequencies) accumulates. Eventually the populations may become so different that they can no longer interbreed to produce fertile offspring: they have become separate species.

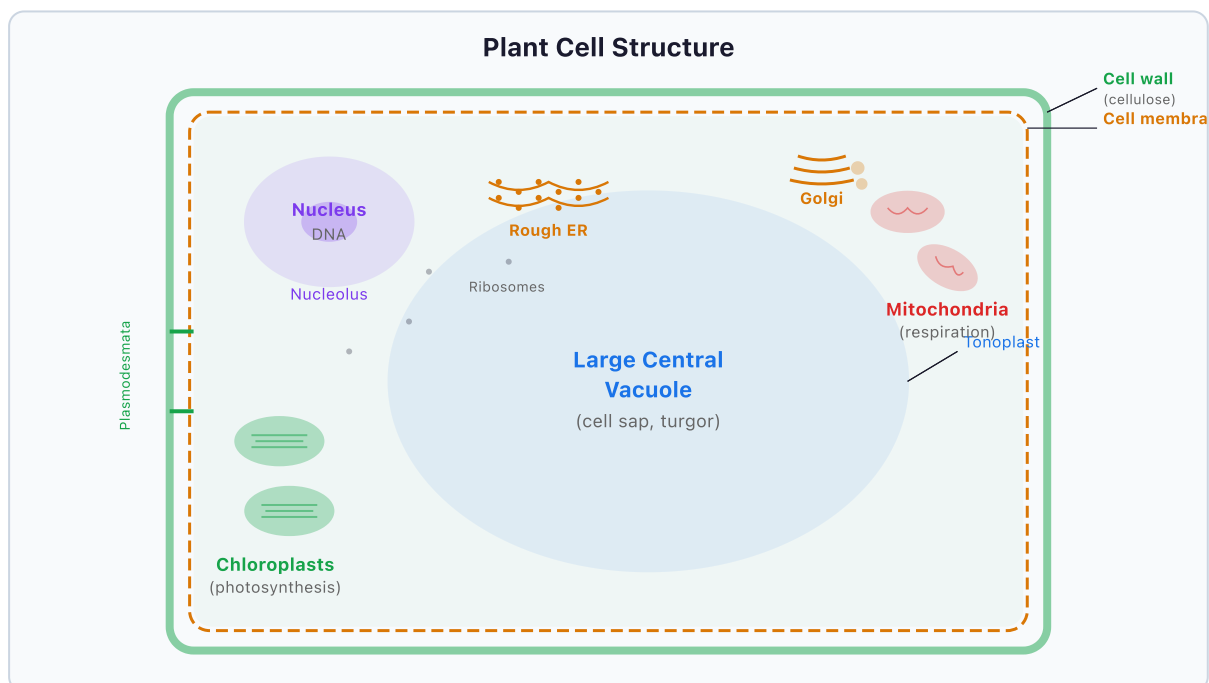
4.3 Plant Cell Biology

Plant Cell Structure

Plant cells share many features with animal cells (nucleus, mitochondria, ribosomes, rough and smooth ER, Golgi apparatus, cell membrane) but also have distinctive features:

- **Cell wall:** Made of cellulose microfibrils embedded in a matrix of hemicellulose and pectins. Provides structural support, prevents the cell from bursting (lysis) when turgid, and is fully permeable to water and dissolved substances.

- **Chloroplasts:** Double membrane-bound organelles containing thylakoid membranes (arranged in grana) and stroma. Site of photosynthesis. Contain their own DNA and 70S ribosomes.
- **Large permanent vacuole:** Membrane-bound (tonoplast), filled with cell sap (water, ions, sugars, pigments). Maintains turgor pressure, supports the cell, and stores substances.
- **Plasmodesmata:** Channels through cell walls connecting the cytoplasm of adjacent cells, allowing transport of molecules and communication between cells.
- **Amyloplasts:** Starch-storing plastids found in storage organs like potato tubers and root cells.
- **Middle lamella:** The layer between the walls of adjacent plant cells, made mainly of calcium pectate, which cements the cells together.
- **Pits:** Thin regions of the cell wall, where little secondary wall material has been laid down; pits in adjacent cells line up, allowing substances to pass between cells.
- **Comparison with animal cells:** Animal cells have no cell wall, middle lamella, pits, plasmodesmata, chloroplasts, amyloplasts or large permanent vacuole with a tonoplast; they have centrioles (absent from most plant cells) and may have small temporary vacuoles. Both have a nucleus, mitochondria, ribosomes, rER, sER, Golgi apparatus and cell surface membrane. You should be able to recognise all of the plant cell organelles listed here in electron micrographs.



Cellulose and Plant Fibres

Cellulose is a polysaccharide made of long, unbranched chains of **beta-glucose** monomers joined by 1,4-glycosidic bonds. Because beta-glucose molecules are oriented alternately (each rotated 180°

relative to its neighbour), the chain is straight rather than coiled. Adjacent chains are cross-linked by extensive **hydrogen bonds**, forming microfibrils of great tensile strength (comparable to steel on a weight-for-weight basis). These microfibrils are embedded in a matrix of hemicellulose and pectins, creating a composite material similar in principle to fibreglass or reinforced concrete.

Starch is the energy store in plants, found in amyloplasts. It is made of α -glucose and is a mixture of two polysaccharides: **amylose** (unbranched chains of α -glucose joined by 1,4-glycosidic bonds, which coil into a compact helix) and **amylopectin** (chains with 1,4-glycosidic bonds and branches formed by 1,6-glycosidic bonds, so there are many ends where enzymes can hydrolyse glucose off rapidly). Starch is compact and insoluble, so it does not affect the water potential of the cell. In contrast, in cellulose every other β -glucose is inverted, forming straight chains; hydrogen bonds form between the $-\text{OH}$ groups of β -glucose molecules in neighbouring parallel chains, bundling 60–70 chains together into a **microfibril**.

Cellulose cannot be digested by most animals because they lack the enzyme **cellulase** — only certain bacteria, fungi, and protists produce it. Ruminant animals (e.g., cows) have mutualistic microorganisms in their rumen that produce cellulase, allowing them to break down plant cell walls. Cellulose is the most abundant organic molecule on Earth, with plants producing an estimated 10^{11} tonnes per year.

Comparing structural polysaccharides: Cellulose (in plant cell walls) and chitin (in fungal cell walls and arthropod exoskeletons) are both structural polysaccharides that are insoluble, strong, and chemically unreactive. Chitin is similar to cellulose but has a nitrogen-containing acetylamine group added to each glucose monomer (N-acetylglucosamine), making it even stronger.

Plant fibres (sclerenchyma and xylem) are strengthened by additional compounds: lignin (waterproofs and adds rigidity) and sometimes suberin. These natural fibres have many human uses including textiles (cotton, flax/linen), rope (hemp, sisal), paper, and building materials.

Microfibril Arrangement and Secondary Thickening

In the cell wall, cellulose microfibrils are laid down in layers, with the microfibrils in each layer at an angle to those in the next (a net-like arrangement). This gives the wall great strength in all directions. In **xylem vessels** and **sclerenchyma fibres**, **secondary thickening** occurs: extra layers of cellulose are laid down inside the primary wall and impregnated with **lignin**. This makes the walls thick, rigid and of high tensile strength, so these fibres are strong and resist breaking, properties exploited by humans in ropes, fabrics and composite materials.

Sclerenchyma, Xylem and Phloem

Feature	Sclerenchyma fibres	Xylem vessels	Phloem sieve tubes
Structure	Long, narrow dead cells with tapering, overlapping end walls; thick lignified walls; hollow lumen	Long hollow tubes of dead cells with end walls broken down; lignified walls with pits	Living sieve tube elements with sieve plates (end walls with pores); little cytoplasm, no nucleus; companion cells alongside; cellulose walls, not lignified
Position in the stem	In the vascular bundles, on the outside of the phloem	In the vascular bundles, on the inside (towards the centre of the stem)	In the vascular bundles, between the xylem and the sclerenchyma
Function	Support	Support, and transport of water and mineral ions	Translocation of organic solutes (e.g. sucrose, amino acids)

Similarities: all three are elongated cells arranged end to end in the vascular bundles; sclerenchyma fibres and xylem vessels are both dead at maturity with lignified secondary walls.

Core Practical 6 (9BN0): identify sclerenchyma fibres, phloem sieve tubes and xylem vessels and their location within stems through a light microscope. Cut thin transverse sections of a stem (e.g. celery) with a sharp blade, stain with toluidine blue O (lignified walls of xylem and sclerenchyma stain blue-green; unligified walls stain pink-purple), mount on a slide and draw a low-power plan showing the positions of the tissues in the vascular bundles.

Core Practical 8 (9BN0): determine the tensile strength of plant fibres. Extract fibres (e.g. by retting stems), clamp one end of a fibre and hang masses from the other, adding them in small increments until the fibre breaks. Keep fibre length and thickness the same and repeat with several fibres. Tensile strength = breaking force / cross-sectional area (force = mass $\times$ 9.81 N kg⁻¹; area = πr^2). For example, a fibre 0.5 mm in diameter breaking at 2.5 kg: F = 24.5 N, A = 1.96×10^{-7} m², tensile strength $\approx 1.2 \times 10^8$ Pa.

4.4 Plant Uses and Sustainability

Plants as a Resource

Humans depend on plants for food (crops), medicines (e.g., aspirin from willow bark, digitalis from foxglove, taxol from yew), fibres, fuels (wood, biofuels), and building materials. Plant breeding and genetic modification have enhanced crop yields, nutritional content, and pest resistance. However, monoculture farming reduces biodiversity and increases vulnerability to disease.

Water and Inorganic Ions

Water is needed by plants as a reactant in photosynthesis, as a solvent and transport medium (in xylem and phloem), to keep cells turgid for support, and for cooling by evaporation in transpiration.

Ion	Why the plant needs it	Deficiency symptoms
Nitrate (NO_3^-)	Nitrogen to make amino acids and proteins, nucleic acids (DNA, RNA), ATP and chlorophyll	Stunted growth; yellowing of the older leaves
Calcium (Ca^{2+})	Forms calcium pectate in the middle lamella, holding cells together; needed in cell membranes	Poor growth; weak, damaged growing points and young leaves
Magnesium (Mg^{2+})	Part of the chlorophyll molecule	Chlorosis (yellowing of leaves, especially between the veins)

Core Practical 7 (9BN0): investigate plant mineral deficiencies. Grow seedlings (e.g. rapid-cycling brassicas or duckweed) in a complete culture solution and in solutions each lacking one ion (nitrate, calcium or magnesium). Control light intensity, temperature and the volume of solution; cover the containers to exclude light and prevent algal growth. Over several weeks record growth (e.g. height, number of leaves, dry mass) and leaf colour, and compare with the complete-solution control.

Drug Discovery from Plants

Many pharmaceuticals are derived from plant secondary metabolites (chemicals produced for defence or signalling, not directly involved in growth). Examples include:

Drug	Plant Source	Medical Use
Aspirin (acetylsalicylic acid)	Willow bark (<i>Salix</i>)	Pain relief, anti-inflammatory, anticoagulant

Digitalis	Foxglove (<i>Digitalis purpurea</i>)	Treatment of heart failure (increases cardiac output)
Taxol (paclitaxel)	Pacific yew (<i>Taxus brevifolia</i>)	Anti-cancer drug (prevents cell division)
Quinine	Cinchona bark (<i>Cinchona</i>)	Anti-malarial
Morphine	Opium poppy (<i>Papaver somniferum</i>)	Pain relief (analgesic)

The Development of Drug Testing

William Withering (1775) learned of a herbal remedy (a “soup” of many plants) used to treat dropsy (fluid build-up caused by heart failure). He identified foxglove as the active ingredient and tested different doses of dried foxglove leaf on his patients, recording the effects. By trial and error he found a dose that was effective without being fatal. His method was not a controlled trial: there was no control group, the patients knew they were being treated, and the amount of active drug varied between leaves, so some patients were poisoned.

Contemporary protocols: After preclinical testing on cells, tissues and animals, drugs go through **three-phased testing**:

- **Phase 1:** A small number (e.g. 20–100) of healthy volunteers; tests safety, side effects and how the body handles different doses.
- **Phase 2:** A larger number (e.g. 100–500) of patients with the condition; tests whether the drug works and finds the best dose.
- **Phase 3:** Thousands of patients; compares the drug with a placebo or the existing treatment to confirm effectiveness and detect rarer side effects.

A **placebo** is an inactive substance that looks identical to the drug; it is used as a control to rule out the placebo effect (patients improving because they believe they are being treated). In a **double blind trial**, patients are randomly allocated to groups and neither the patients nor the doctors know who is receiving the drug and who the placebo, so their expectations cannot bias the results.

The destruction of biodiversity-rich habitats (such as tropical rainforests) may result in the loss of undiscovered medicinal compounds. This provides an economic argument for conservation alongside ecological and ethical arguments.

Conditions for Bacterial Growth

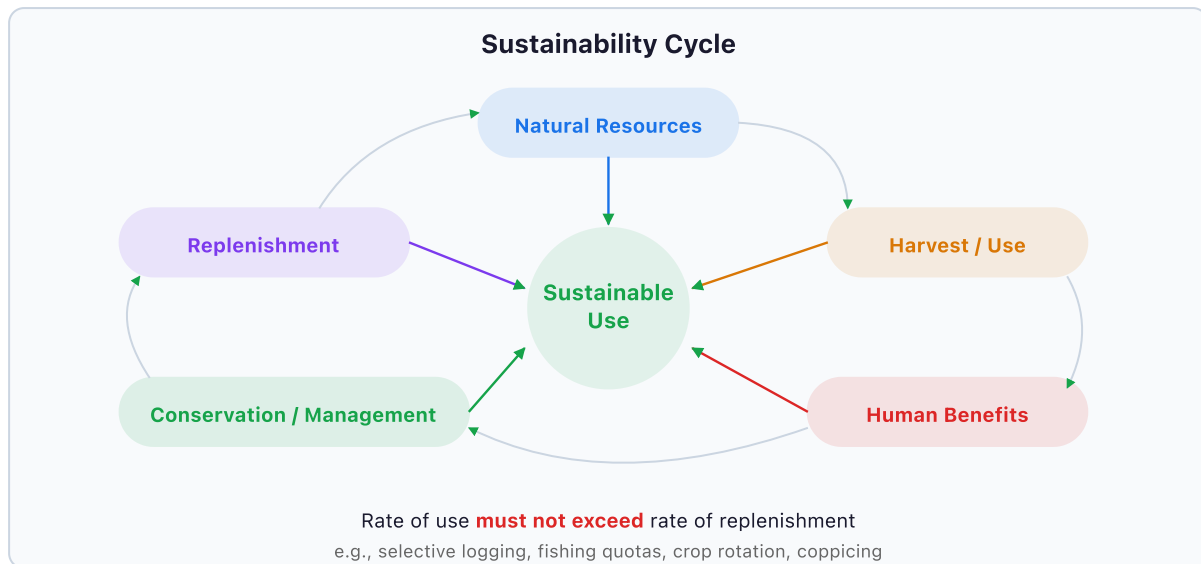
Bacteria need: **nutrients** (a carbon source such as glucose, a nitrogen source for making proteins, and mineral ions), **water**, a suitable **temperature** (many human pathogens grow best at about 37 °C), a suitable **pH**, and **oxygen** for aerobic bacteria (obligate anaerobes are killed by it).

Core Practical 9 (9BN0): investigate the antimicrobial properties of plants, including aseptic techniques for the safe handling of bacteria. Crush plant material (e.g. garlic, mint) with ethanol to extract antimicrobial substances and filter. Soak sterile paper discs in the extract, let the ethanol evaporate, and place them on an agar plate spread with a lawn of bacteria; include a control disc soaked only in ethanol (then evaporated). Incubate at 25 °C (not 37 °C, to reduce the risk of growing human pathogens), then measure the diameter of the clear zone of inhibition around each disc; a larger zone indicates a more effective antimicrobial (area = πr^2).

Aseptic techniques: disinfect the bench; work close to a Bunsen flame so air currents carry microorganisms upwards and away; flame the inoculating loop and the necks of bottles; use sterile equipment and media; lift the Petri dish lid only slightly and briefly; secure the lid with two strips of tape (not sealed all round, so aerobic conditions are maintained); do not open plates after incubation; dispose of all cultures by autoclaving.

Sustainability

Sustainability: Meeting the needs of the present generation without compromising the ability of future generations to meet their own needs. In biological terms, this means using resources at a rate that allows them to be replenished naturally.



Sustainable practices include:

- **Sustainable forestry:** Selective logging, replanting, coppicing, and long rotation periods to maintain forest cover while providing timber.
- **Sustainable fishing:** Quotas, mesh size regulations, closed seasons, marine protected areas to allow fish stocks to recover.
- **Crop rotation:** Growing different crops in sequence to maintain soil fertility, break pest cycles, and reduce the need for chemical inputs.
- **Agroforestry:** Combining trees with crops or livestock to improve productivity, biodiversity, and soil quality.

Plant Fibres, Starch and Sustainability

Plant fibres and starch can contribute to sustainability because they are **renewable** (new crops can be grown each year) and **biodegradable**. Uses include:

- **Plant-based plastics (bioplastics):** Starch (e.g. from maize or potatoes) can be made into plastics for packaging and disposable items, replacing oil-based plastics. They break down more readily and reduce the use of non-renewable crude oil.
- **Plant fibres:** Fibres such as hemp, flax and jute are used for ropes, textiles, paper and in composite materials (e.g. car door panels), replacing oil-based synthetic fibres and plastics.

Growing the plants takes up carbon dioxide, so using them releases less net CO₂ than using fossil fuels, and less energy may be needed to make the products. However, the crops need land that could grow food, and energy is still used for fertilisers, harvesting and processing.

4.5 Conservation of Biodiversity

Threats to Biodiversity

The main threats to global biodiversity include:

- **Habitat destruction:** Deforestation, urbanisation, agricultural expansion. The single greatest threat to biodiversity worldwide.
- **Habitat fragmentation:** Breaking continuous habitats into smaller, isolated patches, reducing gene flow between populations and making them more vulnerable to extinction.
- **Overexploitation:** Overhunting, overfishing, overcollection (e.g., of wild plants for traditional medicine).
- **Pollution:** Chemical pollution, eutrophication, plastic pollution, light and noise pollution.
- **Climate change:** Changing temperature and rainfall patterns, rising sea levels, ocean acidification.
- **Invasive species:** Non-native species that outcompete, prey on, or bring diseases to native species.

Conservation Strategies

In-situ conservation: Protecting species in their natural habitat. Examples include national parks, nature reserves, Sites of Special Scientific Interest (SSSIs), wildlife corridors, and marine protected areas. This maintains natural ecological interactions and allows evolution to continue.

Ex-situ conservation: Protecting species outside their natural habitat. Examples include zoos, botanic gardens, seed banks (e.g., Svalbard Global Seed Vault), captive breeding programmes, and frozen zoos (storing genetic material). These are important for critically endangered species but cannot conserve entire ecosystems.

Evaluating Zoos and Seed Banks

Zoos help conserve endangered species and their genetic diversity through:

- **Captive breeding programmes:** Records of each animal's ancestry (studbooks) are kept and animals are exchanged between zoos (or artificial insemination is used) to avoid inbreeding and maintain genetic diversity.
- **Reintroduction programmes:** Animals bred in captivity are released into the wild to restore populations (e.g. Arabian oryx).

- **Scientific research:** Into the species' reproduction, behaviour, nutrition and disease, which supports breeding and conservation in the wild.
- **Education:** Raising public awareness of conservation and raising funds.

Limitations: captive populations are small, so genetic diversity can still be lost; some animals do not breed well in captivity; reintroduced animals may lack survival skills (finding food, avoiding predators), may carry or lack resistance to diseases, and will fail unless the original threats to their habitat have been removed; programmes are expensive.

Seed banks (e.g. the Millennium Seed Bank at Wakehurst) store seeds that have been dried and kept at about -20°C , so they remain viable for many years. Seeds are collected from many plants and populations to capture genetic diversity. Samples are tested periodically for germination and, if viability falls, plants are grown to produce fresh seed. Research is done into the conditions needed for germination and storage. Advantages: large numbers of species and seeds can be stored cheaply in a small space, for a long time, with little risk of disease. Limitations: some seeds (recalcitrant seeds) cannot survive drying or freezing; stored seeds do not evolve in response to changing environmental conditions; seeds collected may not represent all of the species' genetic diversity; reintroduction needs suitable habitat.

Exam Tip: In essay questions about conservation, discuss reasons for conserving biodiversity (ecological, economic, ethical, aesthetic), the advantages and limitations of both in-situ and ex-situ approaches, and suggest how the two can work together (e.g., captive breeding followed by reintroduction to protected habitats).

International Cooperation

Biodiversity conservation requires international agreements because species and ecosystems cross national boundaries. Key agreements include the Convention on Biological Diversity (CBD), CITES (Convention on International Trade in Endangered Species), the Rio Earth Summit targets, and national biodiversity action plans. However, enforcement can be difficult, and economic pressures often conflict with conservation goals in developing nations.